

GABRIEL EZE

0.1 Complex Analysis Basics

A complex number is any number that can be written in the form $z = a + bi$ where a is the real part and b is the imaginary part

The complex conjugate can be written as $\bar{z} = a - bi$

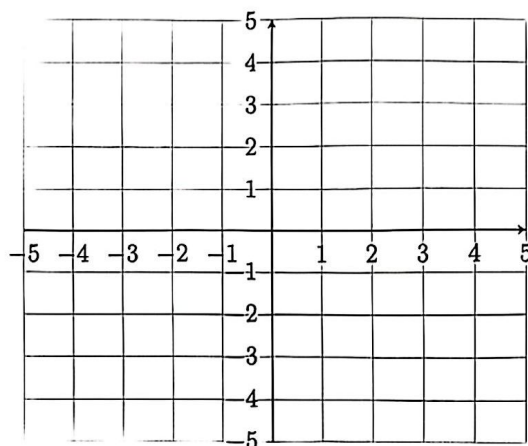
The modulus is the distance from origin written as $|z| = \sqrt{a^2 + b^2}$

To write a complex number in polar form we write it as $z = |z|(\cos \theta + i \sin \theta)$

Example

$$3 + 3i$$

$$\begin{aligned} 1. \text{ Re} &= 3 \\ \text{Im} &= 3 \end{aligned}$$



2. Complex Conjugate

$$\bar{z} = 3 - 3i$$

3. Modulus

$$|z| = \sqrt{9 + 9} = 3\sqrt{2}$$

4. Polar Form

$$3\sqrt{2} \left(\cos\left(\frac{\pi}{4}\right) + i \sin\left(\frac{\pi}{4}\right) \right)$$

0.2 Complex Numbers and Operators

Algebraic:

1. What is $(3 + 4i) + (7 + 2i)$?

$$10 + 6i$$

2. What is $(5 - 2i)(3 + 8i)$?

$$(15 + 16) + (40 - 6)i$$

Geometric: $(a - bi) + (a + bi)i$

$$\text{Let } z_1 = 2(\cos 30^\circ + i \sin 30^\circ), \quad z_2 = 3(\cos 60^\circ + i \sin 60^\circ).$$

1. What is z_1/z_2 ?

$$\frac{2}{3} \cos(-30) + i \sin(-30)$$

2. What is $z_1 - z_2$?

0.3 Euler's Formula

Euler's Formula: $e^{ix} =$

Use Euler's Formula to solve the following:

1. Convert $3e^{5i}$ to $(a + bi)$ form.

$$= 2.852 - 2.817i$$

0.4 Complex Functions

- Definition: Let S be a set of complex numbers. A function f is defined on S as the mapping of each $z \in S \rightarrow w$ where

$$w \in \mathbb{C} \text{ such that } f(z) = w$$

- Example: $f(z) = z^2$. Since z is a complex number, we can write this in the form $z = x + yi$, giving us $f(z) = (x + yi)^2 = (x^2 - y^2) + 2xyi$

- Definition: A complex function $f(z)$ is *differentiable* at a point $z_0 \in \mathbb{C}$ if and only if:

$$f'(z_0) = \lim_{\Delta z \rightarrow 0} \frac{f(z_0 + \Delta z) - f(z_0)}{\Delta z}$$

- Definition: $f(z)$ is analytical or holomorphic on a region R of the complex plane if $f(z)$ is differentiable at each $z_0 \in R$.

Are the following functions analytical functions in the complex plane?

1. $f(z) = z^3$

$$\begin{aligned} \frac{df}{dz} &= \lim_{\Delta z \rightarrow 0} \frac{\Delta f}{\Delta z} = \lim_{h \rightarrow 0} \frac{f(z+h) - f(z)}{h} = \frac{(z+h)^3 - z^3}{h} = \frac{(z+h)(z^2 + 2zh + h^2) - z^3}{h} \\ &= \frac{\cancel{z^3} + 2z^2h + zh^2 + \cancel{z^2h} + 2zh^2 + h^3 - \cancel{z^3}}{h} = \frac{3z^2h + 3zh^2}{h} = 3z^2 + 3zh \end{aligned}$$

2. $f(z) = u(x, y) + v(x, y)i = 2y + xi$

$$\frac{df}{dz} = \lim_{\Delta z \rightarrow 0} \frac{\Delta f}{\Delta z} = \lim_{\Delta z \rightarrow 0} \frac{f(z + \Delta z) - f(z)}{\Delta z} =$$

0.5 Real World Applications

The Fourier Function is:

$$F(\omega) = \int_{-\infty}^{\infty} f(x)e^{-\omega j t} dt$$

1. What is the domain of the input function?

Time is seconds

2. What is the domain of the output from the Fourier Transform?

Frequency in hertz

3. Where is the Fourier Transform used?

In audio & signal analysis

The Schrödinger Equation is:

$$i\hbar \frac{\partial}{\partial t} \Psi(r, t) = H\Psi(r, t)$$

1. What quantum phenomena is explained by the complex numbers in the Schrödinger equation?

The phase factor of what allows for interference as observed in double split experiment.